Building a Robust Sales Data Warehouse using Airflow, dbt, and DuckDB

Modern Data Stack: End-to-End ELT Pipeline

Presented By

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Implementing an ELT methodology resolves data inconsistencies and delivers a scalable Star Schema for BI.

THE CHALLENGE

Sales data was previously scattered across multiple CSV files, suffering from structural inconsistencies and missing values that hindered accurate reporting.

THE GOAL

Design and implement a highly scalable Dimensional Model (Star Schema) tailored specifically for the Business Intelligence team.

OUR APPROACH

We adopted a modern ELT (Extract, Load, Transform) methodology. The entire infrastructure is hosted within a fully containerized environment to ensure consistency across development and production.

A modern, containerized tech stack powers orchestration, storage, and transformation.

Orchestration
Apache Airflow deployed via Astro CLI and Docker handles robust pipeline scheduling.

Data Warehouse
DuckDB serves as the analytical engine, providing lightning-fast OLAP performance.

Transformation
dbt (data build tool) manages modular SQL transformations and automated data testing.

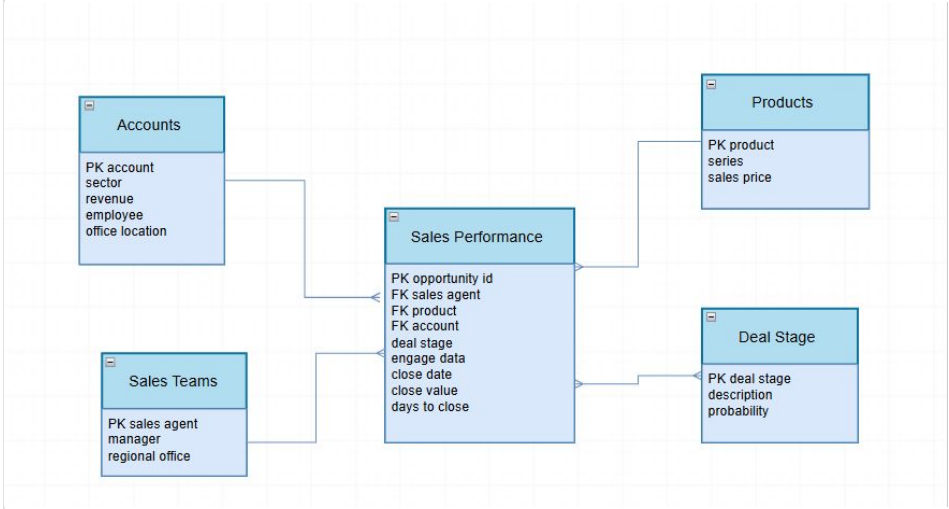
Extraction & Staging
Python and Pandas are utilized for initial data extraction and staging operations.

Version Control
Git and GitHub ensure seamless code management and collaboration.

DATA ARCHITECTURE

Transitioning to a Star Schema optimizes analytical reporting and simplifies downstream queries.

- **Dimensional Model:** We transformed raw, scattered transactional data into a highly structured, business-friendly Star Schema.
- **Fact Table:** fct_sales_performance acts as the central hub for all transactional metrics.
- **Dimension Tables:** dim_accounts, dim_products, and dim_sales_teams provide descriptive context to the sales facts.
- **Business Impact:** This architecture significantly simplifies complex queries, improves data retrieval speeds, and is perfectly optimized for analytical reporting.



The automated Airflow DAG guarantees sequential extraction, transformation, and data quality testing.

EXTRACT & LOAD
Python scripts load CSVs into DuckDB.

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DEPENDENCIES
Executes dbt deps to install testing packages.

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TRANSFORM
dbt run builds the central Fact table.

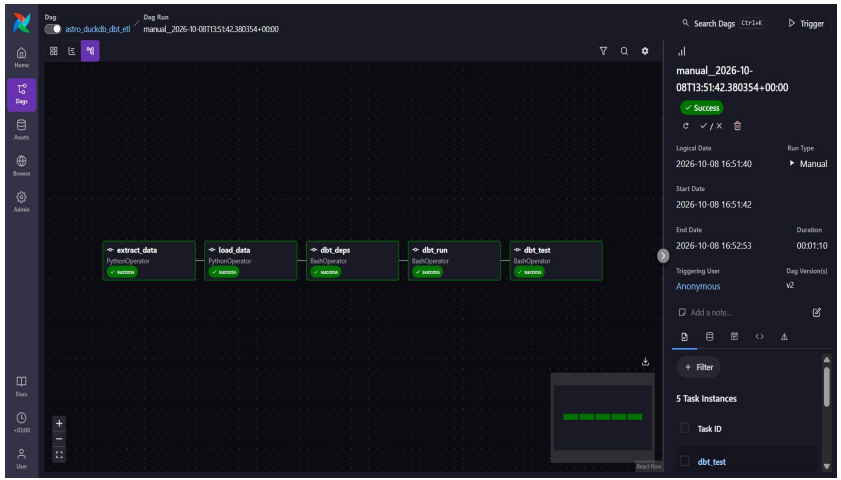
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DATA QUALITY
dbt test validates Unique & Not Null integrity.

Pipeline Operations

- **Extract & Load:** Python scripts systematically read source CSVs and load raw data directly into DuckDB.
- **Dependencies:** The pipeline executes dbt deps to ensure all necessary testing packages are installed.
- **Transform:** dbt run executes staging models and successfully builds the central Fact table.
- **Data Quality:** Finally, dbt test runs rigorous validation checks, ensuring critical fields are Unique, Not Null, and strictly Positive.

Airflow DAG Graph Visualization



MILESTONE & ROADMAP

Achieving 100% data quality compliance paves the way for advanced BI integration and alerting.

KEY ACHIEVEMENT

100%

Data Quality Compliance

We successfully deployed a fully automated ELT pipeline within a containerized infrastructure, ensuring complete structural integrity across all data sources.

NEXT STEPS & FUTURE INITIATIVES

01 / IMMEDIATE

BI Integration

Connecting our DuckDB warehouse to BI tools like Metabase or Power BI for enterprise reporting.

02 / SCALABILITY

Pipeline Optimization

Implementing incremental materialization in dbt to handle growing data volumes efficiently.

03 / RELIABILITY

Proactive Alerting

Integrating automated Slack and Email notifications for instant Airflow DAG failure alerts.

SYSTEM STATUS

PRODUCTION READY

Automated ELT pipeline deployed successfully.